Section 5.4: edits for the portfolio variance bound

The variance penalty weight $(p_i - s_i)^2\sigma_i^2$ is replaced by $(p_i - s_i)^2\sigma_i^2/\mu_i^2$. All other text of Section 5.4 is unchanged.

1. Replace the text from "For the distribution-free setting, the following proposition also establishes ..." (before Proposition 7) through Proposition 7, its proof, and the paragraph "We propose a distribution-free mean-variance newsvendor problem ... for each $i \in N$." by:

Expression (18) depends on the entire distribution F_i. The following bound depends only on (μ_i, σ_i), as does θ_i, and grows with the order quantity.

Lemma 1. *Let $D \geq 0$ have mean $\mu > 0$ and variance σ^2. Then* $\mathrm{Var}(\min\{D, x\}) \leq (\sigma/\mu)^2 x^2$ *for all $x \geq 0$, and the constant $(\sigma/\mu)^2$ cannot be improved.*

Proof. Let $Y = \min\{D, x\}$. We use that for every random variable Y with finite second moment and every constant $c \in \mathbb{R}$,

$$\mathrm{Var}(Y) = E[(Y - E[Y])^2] \leq E[(Y - c)^2], \tag{a}$$

since $E[(Y - c)^2] = \mathrm{Var}(Y) + (E[Y] - c)^2$.

Case $x \geq \mu$. Take $c = \mu$ in (a). If $D \leq x$ then $Y - \mu = D - \mu$; if $D > x$ then $0 \leq x - \mu = Y - \mu < D - \mu$. Hence $|Y - \mu| \leq |D - \mu|$ and

$$\mathrm{Var}(Y) \leq E[(Y - \mu)^2] \leq E[(D - \mu)^2] = \sigma^2 \leq \frac{\sigma^2}{\mu^2} x^2.$$

Case $x < \mu$. Take $c = x$ in (a). Then $(Y - x)^2 = ((x - D)^+)^2$, where $t^+ = \max\{t, 0\}$. If $D < x$ then, since $D \geq 0$ and $x/\mu < 1$, $0 < x - D \leq x - \frac{x}{\mu}D = \frac{x}{\mu}(\mu - D)$; if $D \geq x$ then $(x - D)^+ = 0$. Hence $((x - D)^+)^2 \leq \frac{x^2}{\mu^2}(\mu - D)^2$ for every outcome, and

$$\mathrm{Var}(Y) \leq E\big[((x - D)^+)^2\big] \leq \frac{x^2}{\mu^2} E[(D - \mu)^2] = \frac{\sigma^2}{\mu^2} x^2.$$

Sharpness. For $q \in (0, 1)$ let $P(D = 0) = q$ and $P(D = \mu/(1 - q)) = 1 - q$, so $E[D] = \mu$, and let $x = \mu/(1 - q)$. Then $Y = D$, so $\mathrm{Var}(Y) = \sigma^2$, while the bound equals $\sigma^2/(1 - q)^2$. The ratio $(1 - q)^2$ tends to 1 as $q \to 0$, so no constant smaller than σ^2/μ^2 is valid for all distributions. □

Remark. *Lemma 1 also follows from the Lorenz order: for $g \geq 0$ nondecreasing with $g(t)/t$ non-increasing, $g(D)$ is Lorenz-smaller than D (Fellman, 1976; Arnold, 1987). With $g(t) = \min\{t, x\}$, the coefficient of variation of sales does not exceed σ/μ, so* $\mathrm{Var}(\min\{D, x\}) \leq (\sigma/\mu)^2 \min\{x, \mu\}^2$*; we use the convex bound $(\sigma/\mu)^2 x^2$.*

Since $V[\pi_i(x_i)] = (p_i - s_i)^2 \,\mathrm{Var}(\min\{D_i, x_i\})$, for independent demands Lemma 1 gives

$$V\Big[\sum_{i \in N} \pi_i(x_i)\Big] \leq \sum_{i \in N} (p_i - s_i)^2 \frac{\sigma_i^2}{\mu_i^2} x_i^2 = \mathrm{Var}\Big(\sum_{i \in N} (p_i - s_i)\, x_i \frac{D_i}{\mu_i}\Big), \tag{18′}$$

the Markowitz variance of a portfolio with positions $(p_i - s_i)x_i$ in independent unit-mean assets D_i/μ_i. (For correlated demands, Minkowski's inequality gives the bound $(\sum_i (p_i - s_i)(\sigma_i/\mu_i)x_i)^2$

instead.) We propose a distribution-free mean-variance model that adds $\lambda > 0$ times the bound (18′) to the Gallego–Moon cost bound. Each product's cost $\theta_i(x_i) + \lambda(p_i - s_i)^2(\sigma_i/\mu_i)^2 x_i^2$ is strictly convex, and x_i^U denotes its unconstrained minimizer, the root of a quartic first-order condition.

2. In (19a) and (20) replace $(p_i - s_i)^2\sigma_i^2$ by $(p_i - s_i)^2\sigma_i^2/\mu_i^2$:

$$\text{(19a)} \quad \min_{x,\theta,z} \sum_{i\in N} \theta_i + \lambda \sum_{i\in N} (p_i - s_i)^2 \frac{\sigma_i^2}{\mu_i^2} x_i^2, \qquad \text{(20)} \quad \dots \sum_{i\in N} \theta_i + \lambda \sum_{i\in N} (p_i - s_i)^2 \frac{\sigma_i^2}{\mu_i^2} u_i \ \dots$$

3. In "A data-dependent range for the penalty parameter" update mc_i and $\lambda_{\max}$, and add the last sentence:

$mc_i(x_i) = \partial\theta_i/\partial x_i(x_i) + 2\lambda(p_i - s_i)^2(\sigma_i/\mu_i)^2 x_i$, and

$$\lambda_{\max} = \max_{i\in N:\, \sigma_i > 0} \left[\frac{p_i - s_i}{2} \left(1 + \frac{\mu_i - \varepsilon}{\sqrt{\sigma_i^2 + (\varepsilon - \mu_i)^2}} \right) - (c_i - s_i) \right]_+ \frac{\mu_i^2}{2(p_i - s_i)^2 \sigma_i^2\, \varepsilon}.$$

Because the weights $(\sigma_i/\mu_i)^2$ are dimensionless, λ and $\lambda_{\max}$ do not depend on the unit of demand (with prices per unit).